Nikhilesh Vijay Chirutala

Backend Software Engineer

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SUMMARY

Backend Software Engineer with 3 years of experience designing, developing, and maintaining production applications using Python, FastAPI, REST APIs, MongoDB, PostgreSQL, MySQL, and AWS. Experienced in backend service development, API design, business logic, data processing, cloud storage workflows, third-party integrations, authentication, authorization, and production troubleshooting. Currently developing Python services for a high-volume cloud contact-center platform, working on call-recording processing, recovery, storage, and service integrations. Also experienced in RAG-based applications, LLM integration, CI/CD, observability, and production support.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript (ES6+), TypeScript

Backend & Frameworks: FastAPI, Django, Flask, REST APIs, API Development, Backend Automation

Databases: MongoDB, PostgreSQL, MySQL, SQL, Neon DB

Cloud & AWS: AWS, Amazon S3, AWS Lambda, Vercel

Architecture & Integration: Microservices, WebSockets, API Integration, Third-Party Integrations

AI & LLM: RAG, LLM Integration, Vector Search, Prompt Engineering

Security: Authentication, Authorization, RBAC, ACL, JWT

DevOps & Observability: GitHub Actions, CI/CD, Linux, Datadog, Grafana, Sentry

Testing & Tools: Postman, Jest, Unit Testing, Git, GitHub, Debugging

Frontend: React.js, Next.js, Jinja2, Tailwind CSS

WORK EXPERIENCE

Software Engineer · CA One Tech Cloud Inc. — Apr 2026 – Sep 2026 | Bengaluru, India

Client: Convoso — Cloud Contact Center Platform — *Python, FastAPI, AWS, Amazon S3, MySQL, Datadog*

- Develop and maintain production-grade Python services for a high-volume cloud contact-center platform, supporting call-recording ETL, data processing, storage workflows, and production operations.
- Migrated a legacy Perl-based call-recording workflow to a long-running Python service deployed across 40 production clusters, supporting feature flags, staged rollout, validation, rollback, and monitoring.
- Built Python integrations with a Call Intelligence microservice, implementing service APIs, eligibility checks, account and campaign controls, daily processing limits, retries, response validation, and result persistence.
- Developed validation and recovery workflows for call-recording data, handling incomplete processing, failed operations, and inconsistent storage states across production environments.
- Built AWS S3 audit and recovery workflows for recording storage, including deterministic object handling and idempotent re-upload operations for incomplete or failed storage migrations.
- Supported production deployment and rollout of Python services across multiple clusters, including feature-flag configuration, validation, monitoring, and troubleshooting during migration activities.
- Implemented production observability using Datadog, Grafana, dashboards, and Sentry for monitoring service health, tracking errors, diagnosing processing failures, and supporting production incident resolution.
- Investigated production issues across Python services, APIs, databases, and cloud storage, performing troubleshooting and implementing fixes to maintain reliable recording-processing workflows.

Software Engineer · Mantra Technologies — Oct 2023 – Mar 2026 | Hyderabad, India

- Developed backend services, REST APIs, business logic, and data-driven workflows using Python and FastAPI across multiple production applications.
- Designed and implemented API-driven application functionality, connecting backend services, databases, external integrations, and application workflows.
- Implemented authentication, authorization, and RBAC/ACL-based access-control workflows across application resources.
- Integrated external services and backend components, including communication, storage, notification, and application integrations.
- Supported production releases and resolved application issues through API testing, debugging, root-cause analysis, deployment validation, and defect resolution.

Key Projects:

AI Bull — AI-Powered Trading & Investment Platform — *FastAPI, Python, MongoDB, RAG, LLM, React.js, Next.js*

- Built FastAPI and Python backend APIs and business logic supporting portfolio aggregation, investment analysis, screening, backtesting, and AI-powered portfolio workflows.
- Developed backend workflows for aggregating holdings from multiple Demat accounts and presenting consolidated portfolio information.

- Built portfolio workflows for tracking positions, stocks, and mutual funds, along with detailed investment-level information and reports.
- Designed and implemented a backtesting engine for evaluating trading strategies against historical market data and analyzing strategy performance.
- Developed stock and mutual fund screening workflows with configurable filtering capabilities and detailed asset-level information.
- Implemented a Retrieval-Augmented Generation (RAG) based chatbot that combines portfolio and financial information to provide contextual insights and answer portfolio-related queries.

CRM.io — AI-Driven CRM System *FastAPI, Python, Jinja2, MongoDB, ACL, Twilio, WebSockets, AWS S3, AWS Lambda, FCM*

- Developed backend modules and REST APIs using Python and FastAPI, implementing business logic and workflows across Contacts, Tasks, Deals, Documents, communication, and account management.
- Developed a real-time CRM dialer integrating Twilio and an internal SIP-based telephony application through WebSockets, with AI-generated call scripts provided for relevant calls.
- Implemented CRM-wide soft-delete and Trash workflows with a 30-day retention period, using AWS Lambda for permanent record removal after the retention period.
- Built document management workflows using Amazon S3, including browser-to-S3 client uploads and storage of documents associated with CRM records.
- Developed task cadence workflows for scheduled follow-ups, with Firebase Cloud Messaging used for task notifications.
- Built account-level reporting dashboards and an MCP integration exposing CRM capabilities as tools for AI-driven information retrieval and supported operations.

PERSONAL PROJECTS

Enterprise REST API Platform — Django and Flask Backend Application *Python, Django, Flask, PostgreSQL, REST APIs*

- Built modular REST APIs using Django, Django REST Framework, and Flask for authentication, business logic, and data-driven workflows.
- Integrated PostgreSQL with CRUD operations, database models, relationships, validation, and API error handling.
- Implemented JWT authentication and role-based access control, with API testing and validation using Postman.

AnyScrum — Project Management Platform *Next.js, Neon DB, Clerk, Vercel*

- Developed a project management platform for projects, tasks, and team workflows.
- Implemented authentication and user management with Clerk and persistent application data with Neon DB.

JS Studio — Online JavaScript Playground *Next.js, GitHub API, Vercel*

- Developed an online JavaScript playground for writing and executing JavaScript through a web interface.
- Integrated GitHub APIs and deployed the application with Next.js and Vercel.

EDUCATION

B.Tech — Electrical & Electronics Engineering 2019 – 2023 | GPA: 7.73 / 10

Srinivasa Ramanujan Institute of Technology, Anantapur